ESDU 94031

Issued April 1995
Supersedes in part
ESDU 85033

Increments in aerofoil lift coefficient at zero angle of attack and in maximum lift coefficient due to deployment of a double-slotted or triple-slotted trailing-edge flap, with or without a leading-edge high-lift device, at low speeds

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INCREMENTS IN AEROFOIL LIFT COEFFICIENT AT ZERO ANGLE OF ATTACK AND IN MAXIMUM LIFT COEFFICIENT DUE TO DEPLOYMENT OF A DOUBLE-SLOTTED OR TRIPLE-SLOTTED TRAILING-EDGE FLAP, WITH OR WITHOUT A LEADING-EDGE HIGH-LIFT DEVICE, AT LOW SPEEDS

CONTENTS

INCREMENTS IN AEROFOIL LIFT COEFFICIENT AT ZERO ANGLE OF ATTACK AND IN MAXIMUM LIFT COEFFICIENT DUE TO DEPLOYMENT OF A DOUBLE-SLOTTED OR TRIPLE-SLOTTED TRAILING-EDGE FLAP, WITH OR WITHOUT A LEADING-EDGE HIGH-LIFT DEVICE, AT LOW SPEEDS

1. NOTATION AND UNITS

		SI	*British*
$(a_1)_0$	basic aerofoil lift-curve slope in incompressible flow	rad^{-1}	rad^{-1}
C_{Lm}	maximum lift coefficient of aerofoil with high-lift devices deployed, based on c		
$(C_{LmB})_d$	maximum lift coefficient of basic aerofoil at datum Reynolds number $R_c = 3.5 \times 10^6$		
C_{L0}	lift coefficient at zero angle of attack for aerofoil with high-lift devices deployed, based on c		
ΔC_{Lm}	increment in maximum lift coefficient due to deployment of high-lift devices, based on c, Equation (3.2)		
ΔC_{Lml}	increment in maximum lift coefficient due to deployment of leading-edge high-lift device, based on c, see Section 5		
ΔC_{Lmt}	increment in maximum lift coefficient due to deployment of trailing-edge flap, based on c, Equation (3.4)		
$\Delta C'_{Lmt}$	increment in maximum lift coefficient due to deployment of double-slotted or triple-slotted trailing-edge flap, based on c', at datum Reynolds number $R_c = 3.5 \times 10^6$, Equation (4.19) or (4.31)		
$(\Delta C'_{Lmt})_0$	component of $\Delta C'_{Lmt}$ independent of flap deflection, Equation (4.16)		
$(\Delta C'_{Lmt})_1$	component of $\Delta C'_{Lmt}$ due to flap deflection, Equation (4.16)		
ΔC_{L0}	increment in lift coefficient at zero angle of attack due to deployment of high-lift devices, based on c, Equation (3.1)		
ΔC_{L0l}	increment in lift coefficient at zero angle of attack due to deployment of leading-edge high-lift device, based on c, see Item No. 94027		

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ΔC_{L0t}	increment in lift coefficient at zero angle of attack due to deployment of trailing-edge flap, based on c, Equation (3.3)		
$\Delta C'_{L0t}$	increment in lift coefficient at zero angle of attack due to deployment of double-slotted or triple-slotted trailing-edge flap, based on c', Equation (4.4) or (4.20)		
$\Delta C'_{L0t1}$, $\Delta C'_{L0t2}$	contribution to $\Delta C'_{L0t}$ arising from first, second element of trailing-edge flap, Equation (4.2), (4.3)		
$\Delta C'_{L1}$, $\Delta C'_{L2}$, $\Delta C'_{L3}$	increment in lift coefficient associated with deployment of equivalent first, second, third element of slotted trailing-edge flap on aerofoil with lift-curve slope of 2π, based on c', Figure 4, Figure 5, Figure 6		
c	basic (plain) aerofoil chord (*i.e.* chord with high-lift devices undeployed), see Sketch 4.1 or 4.2	m	ft
c'	extended aerofoil chord (*i.e.* chord with high-lift devices deployed), see Sketch 4.1 and Equation (4.9) for double-slotted flap and Sketch 4.2 and Equation (4.23) for triple-slotted flap	m	ft
c_{et1}, c_{et2}, c_{et3}	equivalent flap chord of first, second, third element of slotted trailing-edge flap, see Sketch 4.1 and Equations (4.13), (4.15) for double-slotted flap, Sketch 4.2 and Equations (4.26), (4.28), (4.30) for triple-slotted flap	m	ft
c_{t1}, c_{t2}, c_{t3}	chord of first, second, third element of slotted trailing-edge flap, see Sketch 4.1 for double-slotted flap, Sketch 4.2 for triple-slotted flap	m	ft
Δc_{t1}, Δc_{t2}, Δc_{t3}	increment in c_{t1}, c_{t2}, c_{t3}, see Sketch 4.1 for double-slotted flap, Sketch 4.2 for triple-slotted flap	m	ft
c'_{t1}, c'_{t2}, c'_{t3}	extended chord of first, second, third element of slotted trailing-edge flap, see Sketch 4.1 and Equations (4.8), (4.11) for double-slotted flap, Sketch 4.2 and Equations (4.8), (4.11) and (4.24) for triple-slotted flap	m	ft
Δc_l	chord extension due to deployment of leading-edge device, see Sketch 4.1 or 4.2	m	ft
F_R	factor for effect of Reynolds number on ΔC_{Lml} and ΔC_{Lmt}, see Equation (3.5)		
J_{t1}, J_{t2}, J_{t3}	correlation factor (efficiency factor) for first, second, third element of slotted trailing-edge flap, Figure 1, Figure 2, Figure 3		
K_l	correlation factor for effect of leading-edge device deflection, see Section 5		

K_T	correlation factor for effect of basic aerofoil geometry, Figure 7		
K_{t1}, K_{t2}, K_{t3}	correlation factor for effect of deflection for first, second, third element of slotted trailing-edge flap, Figure 8, Figure 9		
M	free-stream Mach number		
R_c	Reynolds number based on free-stream conditions and aerofoil chord c		
t	maximum thickness of aerofoil	m	ft
x	chordwise distance aft from basic aerofoil leading edge	m	ft
x_{ts}	chordwise location of flap-shroud trailing edge, see Sketch 4.1 or 4.2	m	ft
x_{um}	chordwise location of maximum upper-surface ordinate	m	ft
$z_{u1.25}$	upper-surface ordinate at $x = 0.0125c$ for basic aerofoil	m	ft
δ_l, δ_l°	deflection of leading-edge device, see Sketch 4.1 or 4.2	rad, deg	rad, deg
δ_{t1}, δ_{t2}, δ_{t3} δ_{t1}°, δ_{t2}°, δ_{t3}°	deflection of first, second, third element of slotted trailing-edge flap, positive trailing edge down, see Sketch 4.1 for double-slotted flap, Sketch 4.2 for triple-slotted flap	rad deg	rad deg
ρ_l	leading-edge radius of basic aerofoil, see Sketch 4.1 or 4.2	m	ft

Subscripts

$(\ )_{expt}$	denotes experimental value
$(\ )_{pred}$	denotes predicted value

2. INTRODUCTION

2.1 Scope of the Item

This Item provides semi-empirical methods for estimating the incremental effects on aerofoil lift at zero angle of attack and on maximum lift due to the deployment of a double-slotted or triple-slotted trailing-edge flap, with or without the deployment of a leading-edge device, at low speeds.

Section 3 summarises the equations relating the contributions to the total lift increments at zero angle of attack and at maximum lift arising from the deployment of leading-edge high-lift devices and trailing-edge flaps. Section 4 presents methods whereby the contributions from double-slotted and triple-slotted trailing-edge flaps are obtained. The contribution to the lift increment at zero angle of attack arising from a variety of leading-edge high-lift devices is obtainable from Item No. 94027 (Derivation 22). However, with regard to the lift increment at maximum lift, the situation is hampered by the lack of test data for other than slats in combination with slotted flaps. Nevertheless, Section 5 provides a guide to the use of the data in Item No. 94027 in the present context for the full range of leading-edge devices covered in that Item.

Section 6 addresses applicability and accuracy, Section 7 gives the Derivation and References and Section 8 presents three detailed examples illustrating the use of the methods.

2.2 Application of Data to Calculation of Total Lift Coefficient Values C_{L0} and C_{Lm}

In order to use the data obtained from the present Item in the wider context in which the total lift coefficient at zero angle of attack, C_{L0}, and at maximum lift, C_{Lm}, are required for an aerofoil with high-lift devices deployed, it is necessary to refer to Item No. 94026 (Reference 26). That Item acts as an introduction to, and a link between, each of the Items in the complete series dealing with the incremental effects of high-lift device deployment on aerofoil lift at zero angle of attack and on maximum lift. It describes how the incremental effects are summed and added to the contributions from the basic (*i.e.* plain) aerofoil to give the total values C_{L0} and C_{Lm}.

3. LIFT COEFFICIENT INCREMENTS ΔC_{L0} AND ΔC_{Lm}

The increments in the lift coefficient at zero angle of attack, ΔC_{L0}, and at maximum lift, ΔC_{Lm}, due to the deployment of a leading-edge high-lift device and a trailing-edge flap on an aerofoil are given by the sum of the individual increments, *i.e.*

$$\Delta C_{L0} = \Delta C_{L0l} + \Delta C_{L0t} \tag{3.1}$$

and

$$\Delta C_{Lm} = \Delta C_{Lml} + \Delta C_{Lmt} . \tag{3.2}$$

The increments in Equations (3.1) and (3.2) are based on the chord, c, of the basic aerofoil.

The values of ΔC_{L0l} and ΔC_{Lml} for various leading-edge devices are obtainable from Item No. 94027. However, see Section 5 with regard to ΔC_{Lml} for use with the present Item.

For correlation purposes it is more convenient to present the right-hand sides of Equations (3.1) and (3.2) in terms of increments based on the extended chord c'. Also, whereas values of ΔC_{L0l} and ΔC_{L0t} can be taken to be independent of Reynolds number, the values of ΔC_{Lml} and ΔC_{Lmt} are influenced by Reynolds number. Thus, for a general trailing-edge flap, Item No. 94028 (Derivation 23) shows that

$$\Delta C_{L0t} = \frac{c'}{c}\Delta C'_{L0t} \tag{3.3}$$

and
$$\Delta C_{Lmt} = F_R \frac{c'}{c}\Delta C'_{Lmt} , \tag{3.4}$$

where
$$F_R = 0.153\log_{10} R_c . \tag{3.5}$$

The value of $\Delta C'_{Lmt}$ in Equation (3.4) is therefore appropriate to the datum Reynolds number $R_c = 3.5 \times 10^6$, for which $F_R = 1$, (see Section 6.1).

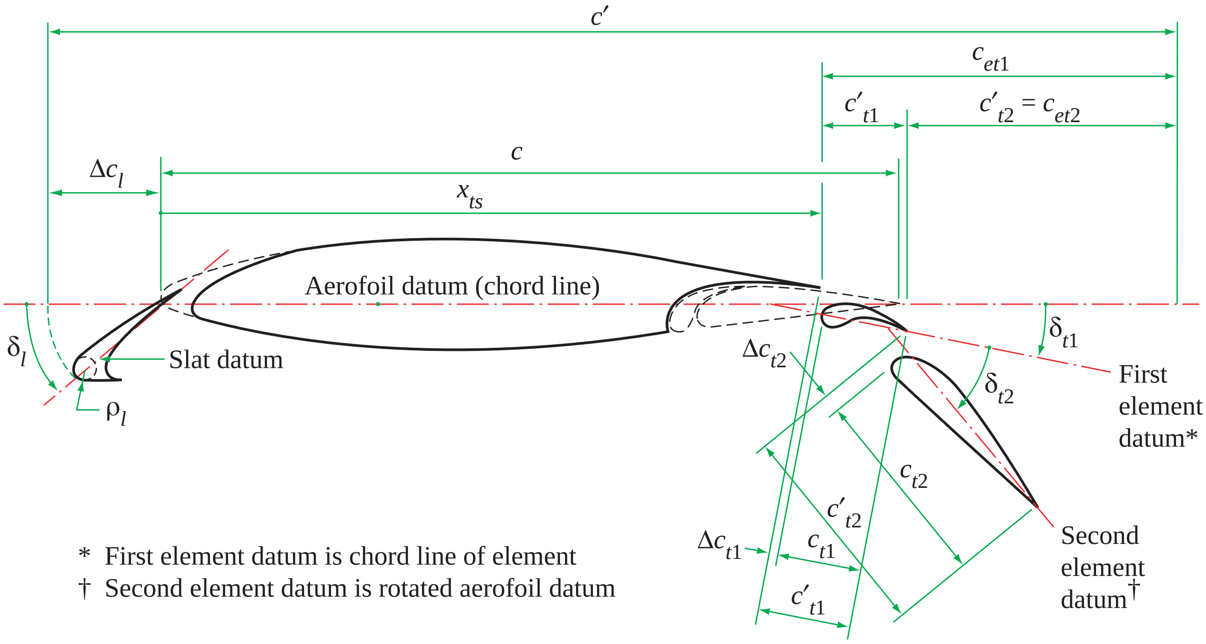
The values of $\Delta C'_{L0t}$ and $\Delta C'_{Lmt}$ in Equations (3.3) and (3.4) are determined for double-slotted and triple-slotted trailing-edge flaps by the methods of Sections 4.1 and 4.2 respectively.

4. LIFT COEFFICIENT INCREMENTS $\Delta C'_{L0t}$ AND $\Delta C'_{Lmt}$ FOR DOUBLE-SLOTTED AND TRIPLE-SLOTTED FLAPS

First approximations to the lift coefficient increments due to the deployment of trailing-edge flaps can be obtained from the theory for an equivalent thin hinged plate with empirical correlation factors to account for the geometry of practical airfoils and flaps. In the theory the lift increment due to a multi-element flap is treated as the sum of the contributions arising from an equivalent flap representing each flap element. To make some allowance for the effects of chord extension in the theory the flap chord ratios and the lift coefficient increments are based on the aerofoil extended chord. This approach was used in Derivation 16 and was the basis for the methods developed for plain trailing-edge flaps in Item No. 94028 which was used as a model for the methods given in the present Item for double-slotted and triple-slotted trailing-edge flaps, see Sections 4.1 and 4.2. However, some adjustments were required to adapt to the considerable departure from the thin hinged plate basis of the model when applied to a slotted flap, possibly involving large extensions in chord.

The presentation of the methods for triple-slotted flaps in Section 4.2 is given in summary only, since the methods of development are identical to those given in detail for double-slotted flaps in Section 4.1.

4.1 Double–slotted Flaps



Sketch 4.1 Double-slotted trailing-edge flap with typical leading-edge high-lift device (slat)

4.1.1 Increment in lift coefficient at zero angle of attack

The increment in lift coefficient at zero angle of attack due to the deployment of a double-slotted trailing-edge flap is given by

$$\Delta C'_{L0t} = \Delta C'_{L0t1} + \Delta C'_{L0t2}, \tag{4.1}$$

where $\Delta C'_{L0t1}$ and $\Delta C'_{L0t2}$ are the contributions arising from the first and second flap elements respectively. Item No. 94030 (Derivation 24) shows that

$$\Delta C'_{L0t1} = J_{t1} \Delta C'_{L1} (a_1)_0 / 2\pi, \tag{4.2}$$

and similarly, $\Delta C'_{L0t2}$ is given by

$$\Delta C'_{L0t2} = J_{t2} \Delta C'_{L2} (a_1)_0 / 2\pi. \tag{4.3}$$

Therefore, substitution of Equations (4.2) and (4.3) into Equation (4.1) gives

$$\boxed{\Delta C'_{L0t} = [J_{t1} \Delta C'_{L1} + J_{t2} \Delta C'_{L2}](a_1)_0 / 2\pi .} \tag{4.4}$$

In this equation J_{t1} and J_{t2} are empirical correlation (or efficiency) factors which are functions of δ_{t1} and are given by

$$J_{t1} = 1.17[\sin(3.83\delta^\circ_{t1})]^{1/2} \qquad \text{for } 0 \le \delta^\circ_{t1} \le 23.5^\circ \tag{4.5}$$

and

$$J_{t1} = 1.17 \qquad \text{for } \delta^\circ_{t1} > 23.5^\circ, \tag{4.6}$$

which are plotted in Figure 1, and

$$J_{t2} = 2.2 - 0.04|\delta^\circ_{t1}| \qquad \text{for } -10^\circ \le \delta^\circ_{t1} \le 30^\circ \tag{4.7}$$

and

$$J_{t2} = 1.0 \qquad \text{for } \delta^\circ_{t1} > 30^\circ, \tag{4.8}$$

which are plotted in Figure 2. *Configurations with negative values of* δ°_{t1} *require special treatment, see Section* 6.1.

The deflection of the first flap element, δ_{t1}, is seen from Sketch 4.1 to be the angle between the aerofoil datum (*i.e.* the aerofoil chord line) and the first element datum. The first element datum (as with the slat) is the chord line of the element, defined as the line passing through the centre of the leading-edge radius and the trailing edge. The deflection of the second flap element, δ_{t2}, is the angle between the first and second element datum lines, see Sketch 4.1. In contrast to the first flap element, the second flap element datum is the rotated aerofoil datum.

The incremental lift coefficients $\Delta C'_{L1}$ and $\Delta C'_{L2}$ are associated with the deployment of the equivalent first and second flap elements respectively on an aerofoil having a lift-curve slope of 2π. They are based on Item Nos Flaps 01.01.08 and Flaps 01.01.09 (Derivations 13 and 12 respectively) and are given in Figures 4 and 5 respectively as functions of δ°_{t1} and c_{et1}/c' for $\Delta C'_{L1}$ and δ°_{t2} and c_{et2}/c' for $\Delta C'_{L2}$.

The parameters $\Delta C'_{L1}$ and $\Delta C'_{L2}$ which are based on test data, replace the value derived from thin plate theory used for plain flaps in Item No. 94028. This was necessary because slot effects are not represented in the simple theory.

The aerofoil extended chord, c', shown in Sketch 4.1 is

$$c' = \Delta c_l + x_{ts} + c'_{t1} + c'_{t2}\ . \tag{4.9}$$

The parameter Δc_l the chord extension due to the deployment of the leading-edge device, obtained for a variety of such devices from Item No. 94027. The quantity x_{ts} is the chordwise location of the trailing edge of the flap shroud and c'_{t1} and c'_{t2} are the extended chord lengths of the first and second flap elements, given by

$$c'_{t1} = c_{t1} + \Delta c_{t1} \tag{4.10}$$

and

$$c'_{t2} = c_{t2} + \Delta c_{t2}\ . \tag{4.11}$$

In Sketch 4.1 Δc_{t1} and Δc_{t2} are drawn as having magnitudes sufficiently small to neglect. For those cases in which Δc_{t1} and Δc_{t2} are not small, graphical estimation would be necessary.

From Sketch 4.1 it is seen that, as used in Derivation 16, the chord lengths of the equivalent flaps representing the first and second flap elements, and required in Figures 4 and 5, are

$$c_{et1} = c'_{t1} + c'_{t2} \tag{4.12}$$

$$= c_{t1} + \Delta c_{t1} + c_{t2} + \Delta c_{t2}, \tag{4.13}$$

from Equations (4.10) and (4.11), and

$$c_{et2} = c'_{t2} \tag{4.14}$$

$$= c_{t2} + \Delta c_{t2} \ . \tag{4.15}$$

The parameter $(a_1)_0$ in Equation (4.4) is the basic aerofoil lift-curve slope in incompressible flow from Item No. Wings 01.01.05 (Derivation 15).

The value of $\Delta C'_{L0t}$ is then used in Equation (3.3) to determine ΔC_{L0t}.

4.1.2 Increment in maximum lift coefficient

In the development of the method for predicting the increment in maximum lift coefficient it was found necessary to adapt further the thin hinged plate theoretical model to cater for slotted flaps, especially those involving large chord extensions. With multi-slotted flaps it is possible to obtain significant increments in maximum lift (with a low drag penalty) by deployment as effectively a single-element flap without deflection. The increment in maximum lift coefficient is then very largely due to chord extension and is related to the basic aerofoil maximum lift coefficient. The increment in maximum lift coefficient can then be considered to consist of two components, one, $(\Delta C'_{Lmt})_0$, independent of flap deflection and the other, $(\Delta C'_{Lmt})_1$, due to the deflection.

Thus

$$\Delta C'_{Lmt} = (\Delta C'_{Lmt})_0 + (\Delta C'_{Lmt})_1, \tag{4.16}$$

where

$$(\Delta C'_{Lmt})_0 = (1 - c/c')(C_{LmB})_d \tag{4.17}$$

and

$$(\Delta C'_{Lmt})_1 = K_T K_{t1} J_{t1} \Delta C'_{L1} + K_T K_{t2} J_{t2} \Delta C'_{L2} - (1 - c/c')\sin\delta_{t1}(C_{LmB})_d, \tag{4.18}$$

so that

$$\boxed{\Delta C'_{Lmt} = (1 - c/c')(1 - \sin\delta_{t1})(C_{LmB})_d + K_T K_{t1} J_{t1} \Delta C'_{L1} + K_T K_{t2} J_{t2} \Delta C'_{L2}\ .} \tag{4.19}$$

In Equation (4.19) $(C_{LmB})_d$ is the maximum lift coefficient for the basic aerofoil at the datum Reynolds number $R_c = 3.5 \times 10^6$, obtained from Item No. 84026 (Derivation 21).

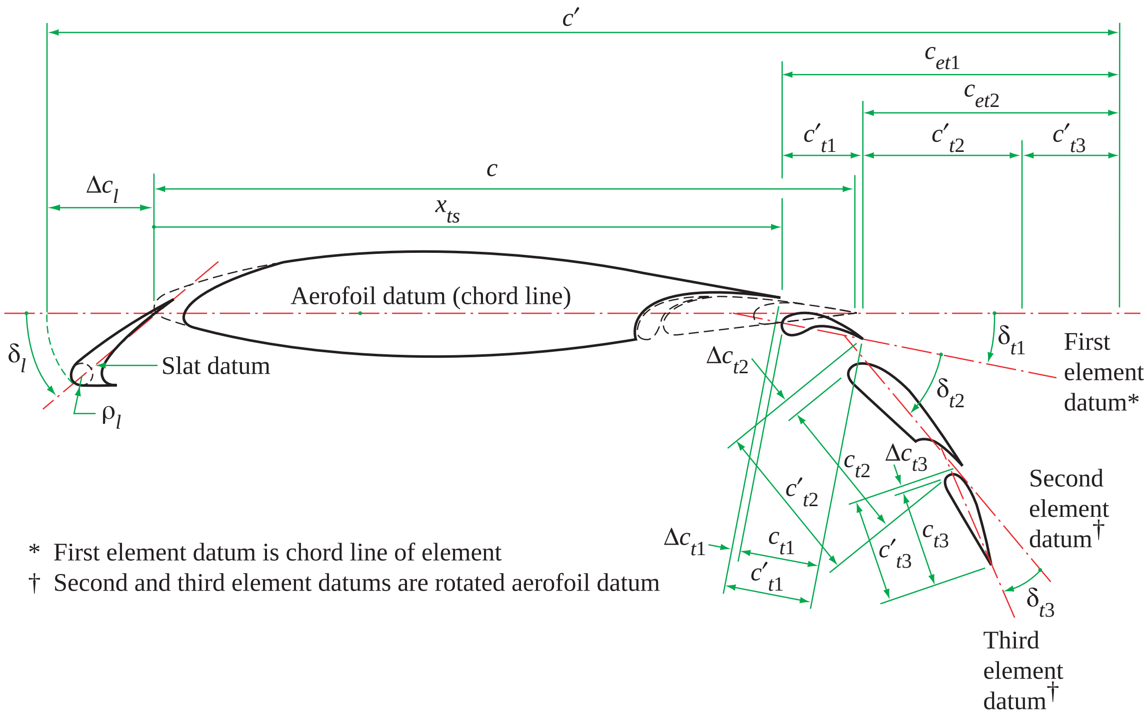
The extended chord, c', is given by Equations (4.9) to (4.11), J_{t1} and J_{t2} are obtained either from Figures 1 and 2 or from Equations (4.5) to (4.8) and $\Delta C'_{L1}$ and $\Delta C'_{L2}$ are obtained from Figures 4 and 5.

The correlation factor K_T, given in Figure 7, allows for the effect of basic aerofoil geometry via $z_{u1.25}/c$, the dimensionless upper-surface ordinate at 0.0125c, and x_{um}/c, the dimensionless chordwise location of the maximum upper-surface ordinate. Values of $z_{u1.25}/c$ for a range of NACA sections are given in Item No. 66034 (Reference 25). The correlation factors K_{t1} and K_{t2} are given as functions of δ°_{t1} in Figures 8 and 9 respectively.

Finally, with the value of $\Delta C'_{Lmt}$ obtained from Equation (4.19), ΔC_{Lmt} is evaluated from Equation (3.4) with F_R given by Equation (3.5).

4.2 Triple-slotted Flaps

The development of the methods for triple-slotted flaps follows very closely that for double-slotted flaps given in Section 4.1 and so the presentation will be given in an abbreviated form.



Sketch 4.2 Triple-slotted trailing-edge flap with typical leading-edge high-lift device (slat)

4.2.1 Increment in lift coefficient at zero angle of attack

The increment in lift coefficient at zero angle of attack due to the deployment of a triple-slotted trailing-edge flap is given by

$$\boxed{\Delta C'_{L0t} = [J_{t1}\Delta C'_{L1} + J_{t2}\Delta C'_{L2} + J_{t3}\Delta C'_{L3}](a_1)_0/2\pi\ .} \qquad (4.20)$$

In this Equation J_{t1} and J_{t2} are obtained either from Figures 1 and 2 or from Equations (4.5) to (4.8) and J_{t3} is obtained either from Figure 3 or the following equations.

$$J_{t3} = 1.42 \qquad \text{for } 0 \le \delta^{\circ}_{t3} < 20^{\circ} \tag{4.21}$$

and $$J_{t3} = 1.42 - 0.004(\delta^{\circ}_{t3} - 20)^{1.79} \qquad \text{for } 20^{\circ} \le \delta^{\circ}_{t3} < 40^{\circ}. \tag{4.22}$$

The parameters $\Delta C'_{L1}$, $\Delta C'_{L2}$ and $\Delta C'_{L3}$ are obtained from Figures 4, 5 and 6 respectively. Figure 6 was obtained from very few data (Derivations 17 and 20) collated assuming similar trends to those for single-slotted and double-slotted flaps in Figures 4 and 5. In order to use Figures 4 to 6 the equivalent flap chord ratios c_{et1}/c', c_{et2}/c' and c_{et3}/c' are required, in which the extended chord c' is given by

$$c' = \Delta c_l + x_{ts} + c'_{t1} + c'_{t2} + c'_{t3} \tag{4.23}$$

where c'_{t1} and c'_{t2} are given by Equations (4.10) and (4.11) respectively and c'_{t3} is given by

$$c'_{t3} = c_{t3} + \Delta c_{t3}, \tag{4.24}$$

see Sketch 4.2. In that sketch Δc_{t1}, Δc_{t2} and Δc_{t3} are drawn as having negligible magnitude. For those cases in which they are not negligible, graphical estimation would be necessary.

From Sketch 4.2 the equivalent flap chords, as used in Derivation 16, are

$$c_{et1} = c'_{t1} + c'_{t2} + c'_{t3} \tag{4.25}$$
$$= c_{t1} + \Delta c_{t1} + c_{t2} + \Delta c_{t2} + c_{t3} + \Delta c_{t3}, \tag{4.26}$$

$$c_{et2} = c'_{t2} + c'_{t3} \tag{4.27}$$
$$= c_{t2} + \Delta c_{t2} + c_{t3} + \Delta c_{t3} \tag{4.28}$$

and

$$c_{et3} = c'_{t3} \tag{4.29}$$
$$= c_{t3} + \Delta c_{t3}. \tag{4.30}$$

The parameter $(a_1)_0$ is obtained from Item No. Wings 01.01.05.

The value of $\Delta C'_{L0t}$ is then used in Equation (3.3) to determine ΔC_{L0t}.

4.2.2 Increment in maximum lift coefficient

The increment in maximum lift coefficient due to the deployment of a triple-slotted trailing-edge flap is given by

$$\boxed{\begin{aligned}\Delta C'_{Lmt} = {} & (1 - c/c')(1 - \sin\delta_{t1})(C_{LmB})_d + K_T K_{t1} J_{t1} \Delta C'_{L1} + K_T K_{t2} J_{t2} \Delta C'_{L2} \\ & + K_T K_{t3} J_{t3} \Delta C'_{L3}\ .\end{aligned}} \tag{4.31}$$

In this equation $(C_{LmB})_d$ is obtained from Item No. 84026 for the basic aerofoil at $R_c = 3.5 \times 10^6$.

The means whereby the extended chord c' and the parameters J_{t1} , J_{t2} , J_{t3} and $\Delta C'_{L1}$, $\Delta C'_{L2}$, $\Delta C'_{L3}$ are obtained are given in Section 4.2.1. The correlation factor K_T is given in Figure 7 and K_{t1} , K_{t2} and K_{t3} are given in Figures 8 and 9.

Finally, with the value of $\Delta C'_{Lmt}$ obtained from Equation (4.31), ΔC_{Lmt} is evaluated from Equation (3.4) with F_R given by Equation (3.5).

5. EFFECT OF LEADING-EDGE HIGH-LIFT DEVICE

The only leading-edge device for which test data are available for aerofoils with slotted trailing-edge flaps is the slat. With regard to the effect of slat deployment on the lift coefficient at zero angle of attack it was found that the increment determined from Item No. 94027 could be used without modification. That being so, it is anticipated that for ΔC_{L0l} the methods of Item No. 94027 for any leading-edge device may be used in conjunction with the present Item for double-slotted or triple-slotted flaps.

Unfortunately, the analysis with regard to maximum lift revealed a more complicated picture. It was found that the optimum slat gap with slotted flaps deployed is larger than for the situation in which the flap is undeployed. The optimum gap is typically within the range 0.02c to 0.05c with slotted flaps deployed, compared with about 0.01c for the flaps undeployed case, for the same range of slat angles. However, only a small variation in ΔC_{Lm} was found within that range, and the test data were used to derive a modified variation of the slat correlation factor K_l with δ_l° compared to those given in Figure 1b of Item No. 94027. The revised factor, which applies only to an optimum (or near-optimum) design, is given in Figure 10. Apart from the value of K_l the method of Item No. 94027 for slats can be assumed to be unchanged.

Since Figure 1b in Item No. 94027 applies to both slats and vented Krüger flaps it is anticipated that Figure 10 here will likewise be applicable to optimised vented Krüger flaps. Further, since the slot flow for the leading-edge device is deemed to be responsible for the changes in K_l , it is felt reasonable to assume that for leading-edge devices with no slot the methods of Item No. 94027 can be used unchanged. However, in the absence of appropriate test data the use of Item No. 94027 for leading-edge devices other than slats in conjunction with the present Item for double-slotted or triple-slotted flaps should be treated with caution in terms of maximum lift.

Finally, it is important to remember that if the leading-edge device extends the aerofoil chord when deployed (*i.e.* $\Delta c_l > 0$, see Equations (4.9) and (4.23)) the effect of that chord extension on c' needs to be taken into account when evaluating Equations (4.4) and (4.19) or (4.20) and (4.31) for double-slotted or triple-slotted trailing-edge flaps respectively.

6. APPLICABILITY AND ACCURACY

6.1 Applicability

Methods are given in this Item for estimating the increments in aerofoil lift coefficient at zero angle of attack and in maximum lift coefficient due to the deployment of a double-slotted or triple-slotted trailing-edge flap with or without the deployment of a leading-edge high-lift device.

Table 6.1 summarises the parameter ranges covered by the measured data, obtained from Derivations 1 to 11, 14 and 17 to 19, from which the various correlation parameters have been obtained. Although test data (Derivations 17 to 19) were available for double-slotted or triple-slotted trailing-edge flaps in conjunction with only one type of leading-edge high-lift device (slat) it is anticipated that, with caution, the methods will apply to any of the types of leading-edge device treated in Item No. 94027, (but see Section 5).

The methods in Section 4 were derived using measured data from wind-tunnel models for which attempts had been made to determine optimum values of the laps and gaps for the slots. The correlations (see Section 6.2) include many data points for which the lap and gap values were near to but not at the optimum, from investigations to find the optimum at each flap deflection. The methods, particularly those for the increment in maximum lift coefficient, therefore apply only to configurations in which the laps and gaps are close to the optimum.

Derivation 1 contains a small number of test data relating to a double-slotted flap with the first element set at $\delta^\circ_{t1} = -10^\circ$ and δ°_{t2} varying. It was found that the values of ΔC_{L0t} and ΔC_{Lmt} in those cases could be predicted quite well (within the scatter bands noted in Section 6.2) if in the methods it was assumed that the first element contributed nothing towards the lift increments, *i.e.* it was assumed that $\Delta C'_{L1} = 0$ in Equations (4.4) and (4.19). On the basis of this limited evidence it is suggested that, with caution, the methods of this Item may be applied, as described, to configurations with first element deflections, δ°_{t1}, down to -10°.

The value $R_c = 3.5 \times 10^6$ was used as the datum from which to develop the factor F_R applicable to the increment in maximum lift coefficient, see Section 3. Most of the data were at or around this value. The effect of Reynolds number on ΔC_{L0t} over the parameter ranges given in Table 6.1 and for higher Reynolds numbers can be assumed to be negligible.

The method of the Item takes no account of Mach number in the increments in maximum lift coefficient due to the deployment of high-lift devices. This is not because such effects are felt to be insignificant, since even at quite low free-stream Mach numbers the local flow around a leading-edge device can attain supersonic velocities. Rather, it is due to the lack of data for free-stream Mach numbers greater than 0.19 for the type of high-lift device considered. The use of the Item is therefore restricted to $M \le 0.2$.

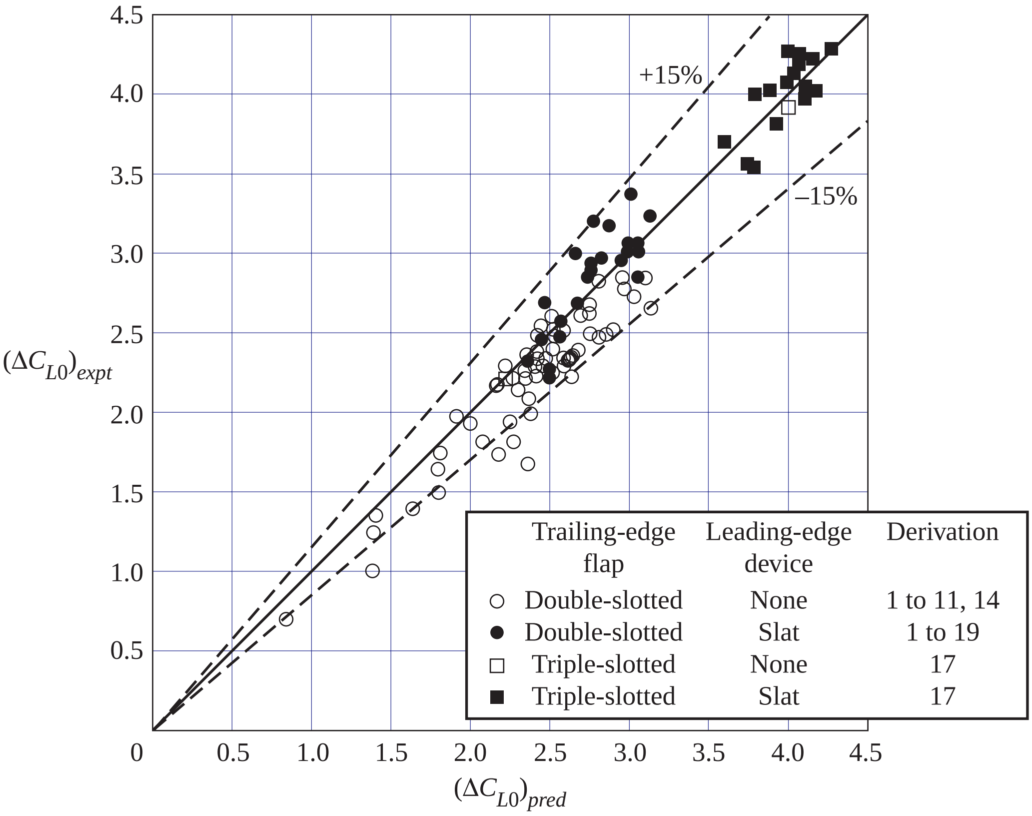
TABLE 6.1 Parameter ranges for test data for double-slotted and triple-slotted trailing-edge flaps used in methods of Section 4

Parameter	*Ranges*	
	Double-slotted flaps	*Triple-slotted flaps*
t/c	0.08 to 0.21	0.15
ρ_l/c	0.004 to 0.049	0.015
$z_{u1.25}/c$	0.011 to 0.049	0.019
x_{um}/c	0.25 to 0.47	0.40
x_{ts}/c	0.715 to 0.854	0.850
c_{t1}/c	0.075 to 0.227	0.131
c_{t2}/c	0.210 to 0.320	0.195 to 0.244
c_{t3}/c	Not applicable	0.161 to 0.210
δ°_{t1}	– 10° to 30°	9° to 19°
δ°_{t2}	10° to 70°	41°
δ°_{t3}	Not applicable	15° to 35°
$\delta^\circ_{t1} + \delta^\circ_{t2}$	10° to 80°	50° to 60°
$\delta^\circ_{t1} + \delta^\circ_{t2} + \delta^\circ_{t3}$	Not applicable	75° to 85°
c'/c (no slat)	1.02 to 1.23	1.34
c'/c (with slat)	1.38 to 1.41	1.43 to 1.47
$R_c \times 10^{-6}$	1.8 to 6.3	1.8 to 3.6
M	0.12 to 0.19	0.12 to 0.19

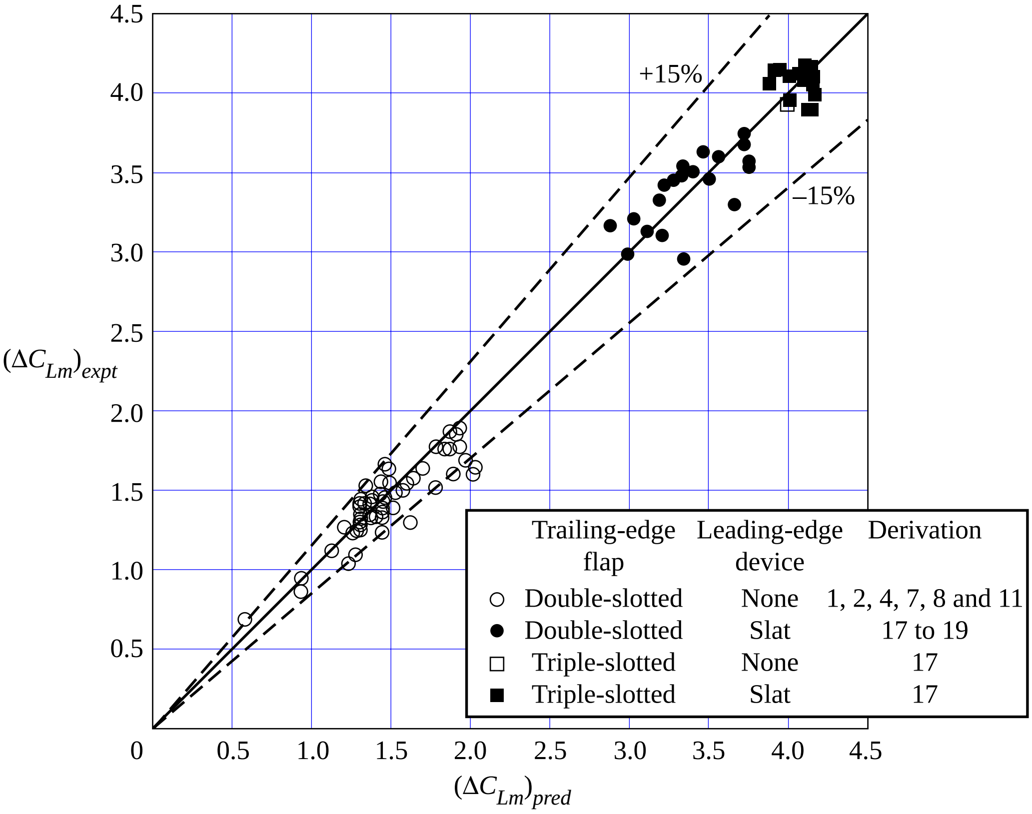
6.2 Accuracy

Sketch 6.1 shows the comparison between predicted and experimental values of ΔC_{L0} due to the deployment of double-slotted or triple-slotted trailing-edge flaps. Also shown are comparisons for double-slotted or triple-slotted flaps in combination with leading-edge slats. With few exceptions the predicted and test data for ΔC_{L0} are correlated to within $\pm 15\%$.

Similarly, Sketch 6.2 presents the corresponding values of the increment in maximum lift coefficient. With few exceptions, the data for ΔC_{Lm} are correlated to within $\pm 15\%$.



Sketch 6.1 Comparison of predicted and experimental values of ΔC_{L0}



Sketch 6.2 Comparison of predicted and experimental values of ΔC_{Lm}

7. DERIVATION AND REFERENCES

7.1 Derivation

The Derivation lists selected sources of information that have assisted in the preparation of this Item.

1.	HARRIS, T.A. RECANT, I.G.	Wind-tunnel investigation of NACA 23012, 23021 and 23030 airfoils equipped with 40-percent-chord double slotted flaps. NACA Rep. 723, 1941.
2.	PURSER, P.E. FISCHEL, J. RIEBE, J.M.	Wind-tunnel investigation of an NACA 23012 airfoil with a 0.30-airfoil-chord double slotted flap. NACA WR L-469, 1943.
3.	BOGDONOFF, S.M.	Wind-tunnel investigation of a low-drag airfoil section with a double slotted flap. NACA WRL-697, 1943.
4.	FISCHEL, J. RIEBE, J.M.	Wind-tunnel investigation of an NACA 23021 airfoil with a 0.32-airfoil-chord double slotted flap. NACA WR L-7, 1944.

5.	CAHILL, J.F.	Aerodynamic data for a wing section of the Republic XF-12 airplane equipped with a double slotted flap. NACA WR L-544, 1946.
6.	QUINN, J.H.	Wind-tunnel investigation of boundary-layer control by suction on the NACA 65_3-418, a = 1.0 airfoil section with a 0.29-airfoil-chord double slotted flap. NACA tech. Note 1071, 1946.
7.	BRASLOW, A.L. LOFTIN, L.K.	Two-dimensional wind-tunnel investigation of an approximately 14-percent-thick NACA 66-series-type airfoil section with a double slotted flap. NACA tech. Note 1110, 1946.
8.	CAHILL, J.F.	Two-dimensional wind-tunnel investigation of four types of high-lift flap on a NACA 65-210 airfoil section. NACA tech. Note 1191, 1947.
9.	QUINN, J.H.	Tests of the NACA 64A212 airfoil section with a slat, a double slotted flap and boundary-layer control by suction. NACA tech. Note 1293, 1947.
10.	QUINN, J.H.	Wind-tunnel investigation of the NACA 65_4-421 airfoil section with a double slotted flap and boundary-layer control by suction. NACA tech. Note 1395, 1947.
11.	CAHILL, J.F. RACISZ, S.F.	Wind-tunnel investigation of seven thin airfoil sections to determine optimum double-slotted-flap configurations. NACA tech. Note 1545, 1947.
12.	ESDU	Lift coefficient increment due to full-span double-slotted flap (main flap slotted). ESDU International, Item No. Flaps 01.01.09, 1948.
13.	ESDU	Lift coefficient increment due to full-span slotted flaps. ESDU International, Item No. Flaps 01.01.08, 1949.
14.	KELLY, J.A. HAYTER, N.-L.F.	Lift and pitching moment at low speeds of the NACA 64A010 airfoil section equipped with various combinations of a leading-edge slat, leading-edge flap, split flap and double-slotted flap. NACA tech. Note 3007, 1953.
15.	ESDU	Slope of lift curve for two-dimensional flow. ESDU International, Item No. Wings 01.01.05, 1955.
16.	SCHEMENSKY, R.T.	Development of an empirically based computer program to predict the aerodynamic characteristics of aircraft. Volume 1: Empirical methods. Convair Aerospace Division, General Dynamics Corporation. AFFDL TR-73-144 (AD 780 100), 1973.
17.	LJUNGSTROM, B.L.G.	2-D wind-tunnel experiments with double and triple slotted flaps. Aeronaut. Res. Inst., Sweden. FFA AU-993, 1973.
18.	LJUNGSTROM, B.L.G.	Two-dimensional wind-tunnel experiments with single and double slotted flaps. Aeronaut. Res. Inst., Sweden. FFA AU-1083, 1975.

19. LJUNGSTROM, B.L.G. Wind-tunnel high lift optimisation of a multiple element airfoil. Aeronaut. Res. Inst., Sweden. FFA AU-778, 1976.

20. BAe, Weybridge Unpublished wind-tunnel test data, 1970 to 1980.

21. ESDU Aerofoil maximum lift coefficient for Mach numbers up to 0.4. ESDU International, Item No. 84026, 1984.

22. ESDU Increments in aerofoil lift coefficient at zero angle of attack and in maximum lift coefficient due to deployment of various leading-edge high-lift devices at low speeds. ESDU International, Item No. 94027, 1994.

23. ESDU Increments in aerofoil lift coefficient at zero angle of attack and in maximum lift coefficient due to deployment of a plain trailing-edge flap, with or without a leading-edge high-lift device, at low speeds. ESDU International, Item No. 94028, 1994.

24. ESDU Increments in aerofoil lift coefficient at zero angle of attack and in maximum lift coefficient due to deployment of a single-slotted trailing-edge flap, with or without a leading-edge high-lift device, at low speeds. ESDU International, Item No. 94030, 1995.

7.2 References

The References list selected sources of information supplementary to that given in this Item.

25. ESDU The low-speed stalling characteristics of aerodynamically smooth aerofoils. ESDU International, Item No. 66034, 1966.

26. ESDU Introduction to the estimation of the lift coefficients at zero angle of attack and at maximum lift for aerofoils with high-lift devices at low speeds. ESDU International, Item No. 94026, 1994.

8. EXAMPLES

8.1 Example 1: Double-slotted Flap

The incremental effects on the lift coefficient at zero angle of attack and on the maximum lift coefficient are to be estimated for the double-slotted flap deployed on a modified smooth NACA 65_2-215 aerofoil as shown in Sketch 8.1. The modification produced a linear profile rearward from 75% chord and 65% chord on the upper and lower surfaces respectively.

The relevant geometrical data are

Aerofoil	Flap first element	Flap second element
$t/c = 0.15$	$c_{t1} = 0.3075$ ft	$c_{t2} = 0.73$ ft
$c = 2.5$ ft	$\delta^\circ_{t1} = 10°$	$\delta^\circ_{t2} = 40°$
$z_{u1.25}/c = 0.0188$	$\Delta c_{t1} = 0$	$\Delta c_{t2} = 0$
$x_{um}/c = 0.40$	$x_{ts} = 2.25$ ft	

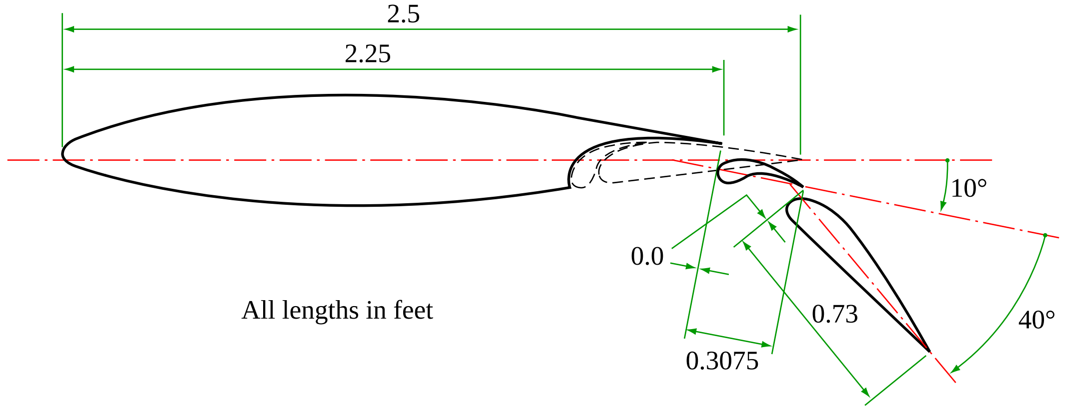
The flow conditions are

$$M = 0.2 \text{ and } R_c = 3.5 \times 10^6 .$$

Also given for the modified NACA 65_2-215 are

$(a_1)_0 = 5.62 \text{ rad}^{-1}$ from Item No. Wings 01.01.05 for boundary layer transition at the leading edge

and $(C_{LmB})_d = 1.309$ from Item No. 84026 for a smooth aerofoil surface at $R_c = 3.5 \times 10^6$.



Sketch 8.1

The extended chords of the first and second flap elements are given by Equations (4.10) and (4.11), *i.e.*

$$\begin{aligned} c'_{t1} &= c_{t1} + \Delta c_{t1} \\ &= 0.3075 + 0 \\ &= 0.3075 \text{ ft} \end{aligned}$$

and

$$\begin{aligned} c'_{t2} &= c_{t2} + \Delta c_{t2} \\ &= 0.73 + 0 \\ &= 0.73 \text{ ft.} \end{aligned}$$

The aerofoil extended chord c' is given by Equation (4.9) as

$$c' = \Delta c_l + x_{ts} + c'_{t1} + c'_{t2}$$

and since there is no leading-edge device $\Delta c_l = 0$, so that

$$\begin{aligned} c' &= 0 + 2.25 + 0.3075 + 0.73 \\ &= 3.2875 \text{ ft.} \end{aligned}$$

Therefore

$$c'/c = 3.2875/2.5 = 1.315\,.$$

Therefore, the equivalent chords of the first and second flap elements are given by Equations (4.12) and (4.14) as

$$\begin{aligned} c_{et1} &= c'_{t1} + c'_{t2} \\ &= 0.3075 + 0.73 \\ &= 1.0375 \text{ ft} \end{aligned}$$

and

$$\begin{aligned} c_{et2} &= c'_{t2} \\ &= 0.73 \text{ ft .} \end{aligned}$$

Therefore, the equivalent chord ratios are

$$\begin{aligned} c_{et1}/c' &= 1.0375/3.2875 \\ &= 0.316 \end{aligned}$$

and

$$c_{et2}/c' = 0.73/3.2875$$
$$= 0.222.$$

Calculation of ΔC_{L0t}:

From Equation (4.4)

$$\Delta C'_{L0t} = [J_{t1}\Delta C'_{L1} + J_{t2}\Delta C'_{L2}](a_1)_0/2\pi.$$

From Equation (4.5) with $\delta^\circ_{t1} = 10°$

$$J_{t1} = 1.17[\sin(3.83\,\delta^\circ_{t1})]^{½}$$
$$= 1.17 \times [\sin(3.83 \times 10°)]^{½}$$
$$= 0.921\,.$$

From Figure 4 with $\delta^\circ_{t1} = 10°$ and $c_{et1}/c' = 0.316$

$$\Delta C'_{L1} = 0.665.$$

From Equation (4.7) with $\delta^\circ_{t1} = 10°$

$$J_{t2} = 2.2 - 0.04|\delta^\circ_{t1}|$$
$$= 2.2 - 0.04 \times 10$$
$$= 1.8.$$

From Figure 5 with $\delta^\circ_{t2} = 40°$ and $c_{et2}/c' = 0.222$

$$\Delta C'_{L2} = 0.940.$$

Therefore, with these values and $(a_1)_0 = 5.62\ \text{rad}^{-1}$, Equation (4.4) is evaluated to give

$$\Delta C'_{L0t} = [0.921 \times 0.665 + 1.8 \times 0.940] \times 5.62/2\pi$$
$$= 2.061,$$

which, in Equation (3.3) gives, with $c'/c = 1.315$,

$$\Delta C_{L0t} = \frac{c'}{c}\Delta C'_{L0t}$$
$$= 1.315 \times 2.061$$
$$= 2.71.$$

Calculation of ΔC_{Lmt}:

From Figure 7 with $z_{u1.25}/c = 0.0188$ and $x_{um}/c = 0.40$

$$K_T = 2.5$$

and from Figures 8 and 9 with $\delta^\circ_{t1} = 10°$

$$K_{t1} = 0.528$$

and $$K_{t2} = 0.215\,.$$

Therefore, Equation (4.19) gives, with these and the other parameter values previously determined, together with $c'/c = 1.315$ and $(C_{LmB})_d = 1.309$,

$$\begin{aligned}\Delta C'_{Lmt} &= (1 - c/c')(1 - \sin\delta_{t1})(C_{LmB})_d + K_T\,K_{t1}\,J_{t1}\,\Delta C'_{L1} + K_T\,K_{t2}\,J_{t2}\,\Delta C'_{L2}\\ &= (1 - 1/1.315) \times (1 - \sin 10°) \times 1.309 + 2.5 \times 0.528 \times 0.921 \times 0.665\\ &\qquad + 2.5 \times 0.215 \times 1.8 \times 0.940\\ &= 0.260 + 0.808 + 0.909\\ &= 1.977\,.\end{aligned}$$

Equation (3.5) gives, for $R_c = 3.5 \times 10^6$,

$$\begin{aligned}F_R &= 0.153\log_{10}R_c\\ &= 0.153 \times \log_{10}(3.5 \times 10^6)\\ &= 1.00\,.\end{aligned}$$

Therefore, Equation (3.4) gives

$$\begin{aligned}\Delta C_{Lmt} &= F_R(c'/c)\Delta C'_{Lmt}\\ &= 1.00 \times 1.315 \times 1.977\\ &= 2.60\,.\end{aligned}$$

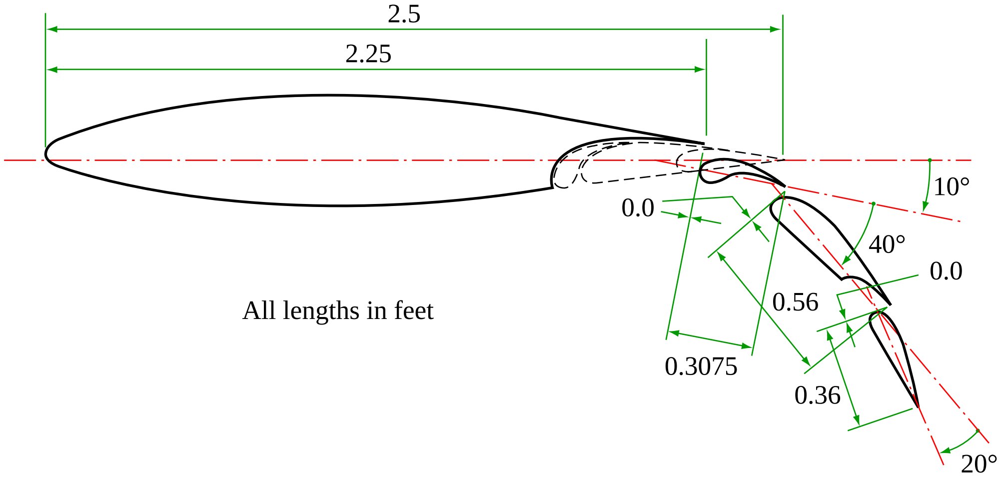
8.2 Example 2: Triple-slotted Flap

The incremental effects on the lift coefficient at zero angle of attack and on the maximum lift coefficient are to be estimated for the triple-slotted flap deployed on the modified smooth NACA 65_2-215 aerofoil of Example 1 as shown in Sketch 8.2. The geometry is the same as that for Example 1 except that the second element has been divided into two elements. The flow conditions and basic aerofoil data are also as for Example 1. Because of the commonality with Example 1 only the flap geometrical data will be given.

Flap first element	Flap second element	Flap third element
c_{t1} = 0.3075 ft	c_{t2} = 0.56 ft	c_{t3} = 0.36 ft
δ°_{t1} = 10°	δ°_{t2} = 40°	δ°_{t3} = 20°

$\Delta c_{t1} = 0$ $\qquad$ $\Delta c_{t2} = 0$ $\qquad$ $\Delta c_{t3} = 0$

$x_{ts} = 2.25$ ft



Sketch 8.2

The form of the calculations follows that used for the double-slotted flap of Example 1.

The extended chords of the three flap elements are obtained using Equations (4.10), (4.11) and (4.24), and since $\Delta c_{t1} = \Delta c_{t2} = \Delta c_{t3} = 0$, they give

$$c'_{t1} = c_{t1} = 0.3075 \text{ ft}$$

$$c'_{t2} = c_{t2} = 0.56 \text{ ft}$$

$$c'_{t3} = c_{t3} = 0.36 \text{ ft.}$$

The extended chord of the aerofoil is given by Equation (4.23) with $\Delta c_l = 0$, since there is no leading-edge device, *i.e.*

$$\begin{aligned} c' &= x_{ts} + c'_{t1} + c'_{t2} + c'_{t3} \\ &= 2.25 + 0.3075 + 0.56 + 0.36 \\ &= 3.4775 \text{ ft,} \end{aligned}$$

so that

$$\begin{aligned} c'/c &= 3.4775/2.5 \\ &= 1.391\,. \end{aligned}$$

The equivalent chords of the three flap elements are obtained using Equations (4.25), (4.27) and (4.29) as

$$\begin{aligned} c_{et1} &= c'_{t1} + c'_{t2} + c'_{t3} \\ &= 0.3075 + 0.56 + 0.36 \\ &= 1.2275 \text{ ft}, \\ c_{et2} &= c'_{t2} + c'_{t3} \\ &= 0.56 + 0.36 \\ &= 0.92 \text{ ft} \end{aligned}$$

and

$$\begin{aligned} c_{et3} &= c'_{t3} \\ &= 0.36 \text{ ft}, \end{aligned}$$

giving the corresponding chord ratios

$$\begin{aligned} c_{et1}/c' &= 1.2275/3.4775 \\ &= 0.353, \\ c_{et2}/c' &= 0.92/3.4775 \\ &= 0.265 \end{aligned}$$

and

$$\begin{aligned} c_{et3}/c' &= 0.36/3.4775 \\ &= 0.104. \end{aligned}$$

Calculation of ΔC_{L0t}:

The values of J_{t1} and J_{t2} are as for Example 1, *i.e.*

$$J_{t1} = 0.921 \text{ and } J_{t2} = 1.8 .$$

Equation (4.22) with $\delta^\circ_{t3} = 20°$ gives

$$J_{t3} = 1.42 .$$

The values of $\Delta C'_{L1}$, $\Delta C'_{L2}$ and $\Delta C'_{L3}$ are obtained from Figures 4 to 6 respectively, which with

	$\delta^\circ_{t1} = 10°$,	$c_{et1}/c' = 0.353$
	$\delta^\circ_{t2} = 40°$,	$c_{et2}/c' = 0.265$
and	$\delta^\circ_{t3} = 20°$,	$c_{et3}/c' = 0.104$
give	$\Delta C'_{L1} = 0.70$,	

$$\Delta C'_{L2} = 1.01$$

and $$\Delta C'_{L3} = 0.347.$$

Therefore, with these values and $(a_1)_0 = 5.62$ rad^{-1} (from Example 1), Equation (4.20) gives

$$\begin{aligned}\Delta C'_{L0t} &= [J_{t1}\,\Delta C'_{L1} + J_{t2}\,\Delta C'_{L2} + J_{t3}\,\Delta C'_{L3}](a_1)_0/2\pi \\ &= [0.921 \times 0.70 + 1.8 \times 1.01 + 1.42 \times 0.347] \times 5.62/2\pi \\ &= 2.643,\end{aligned}$$

which in Equation (3.3), with $c'/c = 1.391$, gives

$$\begin{aligned}\Delta C_{L0t} &= \frac{c'}{c}\Delta C'_{L0t} \\ &= 1.391 \times 2.643 \\ &= 3.68.\end{aligned}$$

Calculation of ΔC_{Lmt}:

From Example 1, $K_T = 2.5$

and from Figures 8 and 9, with $\delta^\circ_{t1} = 10°$,

$$K_{t1} = 0.528,$$
$$K_{t2} = 0.215$$

and $K_{t3} = 0.08$ from Figure 9 with $\delta^\circ_{t3} = 20°$.

Therefore, Equation (4.31) gives, with these and the other parameter values previously determined, together with $c'/c = 1.391$ and $(C_{LmB})_d = 1.309$ (from Example 1)

$$\begin{aligned}\Delta C'_{Lmt} &= (1 - c/c')(1 - \sin\delta_{t1})(C_{LmB})_d + K_T K_{t1} J_{t1}\,\Delta C'_{L1} + K_T K_{t2} J_{t2}\,\Delta C'_{L2} \\ &\qquad + K_T K_{t3} J_{t3} \Delta C'_{L3} \\ &= (1 - 1/1.391) \times (1 - \sin 10°) \times 1.309 + 2.5 \times 0.528 \times 0.921 \times 0.70 \\ &\qquad + 2.5 \times 0.215 \times 1.8 \times 1.01 + 2.5 \times 0.08 \times 1.42 \times 0.347 \\ &= 0.304 + 0.851 + 0.977 + 0.099 \\ &= 2.231,\end{aligned}$$

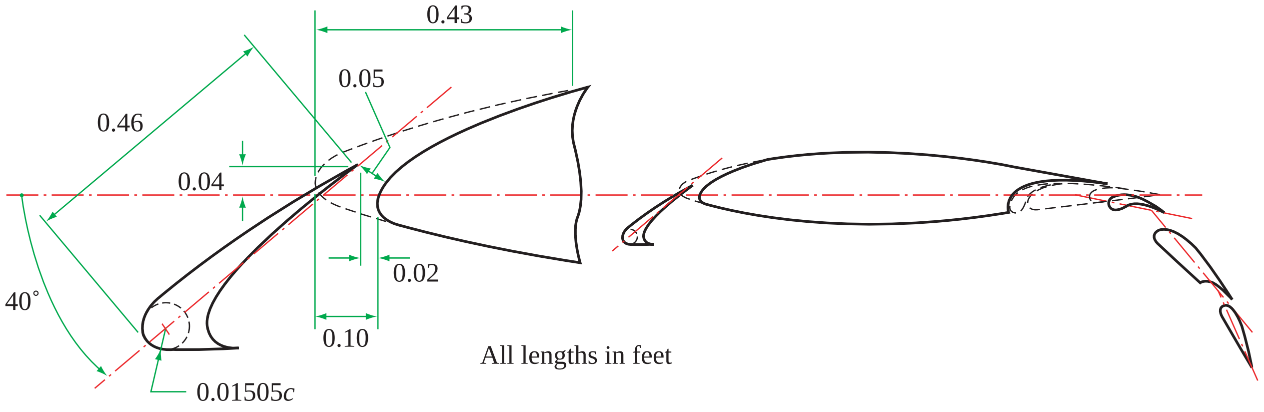
which, in Equation (3.4) with $F_R = 1$ (from Example 1) and $c'/c = 1.391$, gives

$$\begin{aligned}\Delta C_{Lmt} &= F_R(c'/c)\Delta C'_{Lmt} \\ &= 1 \times 1.391 \times 2.231 \\ &= 3.10.\end{aligned}$$

8.3 Example 3: Triple-slotted Flap with Leading-edge Slat

Estimate the effects on the lift coefficient at zero angle of attack and on the maximum lift coefficient of the addition of an optimised leading-edge slat to the triple-slotted flap configuration considered in Example 2, as shown in Sketch 8.3. The relevant geometrical data for the slat (using the notation of Item No. 94027) are:

c_l	$= 0.46$ ft	ρ_l/c	$= 0.01505$
δ_l°	$= 40°$ ($\delta_l = 0.698$ rad)	H_l	$= 0.04$ ft
x_n	$= 0.10$ ft	G_l	$= 0.05$ ft
x_l	$= 0.43$ ft	L_l	$= -0.02$ ft



Sketch 8.3

1. Slotted-Flap Contributions

As noted in Section 5 one effect of the deployed slat is to increase the extended chord c', with the result that the calculations for ΔC_{L0t} and ΔC_{Lmt} from Example 2 will need to be revised for the changed values of c_{et1}/c', c_{et2}/c', c_{et3}/c' and c'/c due to the changed c'.

From Item No. 94027 Equation (4.4a), the chord extension due to slat deployment is

$$
\begin{aligned}
\Delta c_l &= c_l - x_n - L_l - H_l \tan(\delta_l^\circ/2) \\
&= 0.46 - 0.10 - (-0.02) - 0.04 \times \tan(40°/2) \\
&= 0.365 \text{ ft.}
\end{aligned}
$$

The revised value of c' is obtained via Equation (4.23), which indicates that the value of c' from Example 2 (*i.e.* 3.4775 ft) is simply increased by Δc_l, so that

$$
\begin{aligned}
c' &= 0.365 + 3.4775 \\
&= 3.8425 \text{ ft,}
\end{aligned}
$$

giving

$$c'/c = 3.8425/2.5$$
$$= 1.537.$$

The correspondingly changed values of the equivalent chord ratios (with the c_{et1}, c_{et2} and c_{et3} values from Example 2) are

$$c_{et1}/c' = 1.2275/3.8425$$
$$= 0.319,$$
$$c_{et2}/c' = 0.92/3.8425$$
$$= 0.239$$

and

$$c_{et3}/c' = 0.36/3.8425$$
$$= 0.094.$$

From Item No. 94027 (in the notation of that Item), the effective slat chord is,

$$c_{el}/c' = c_l/c' = 0.46/3.8425$$
$$= 0.120.$$

(i) *Contribution to* ΔC_{L0}:

These revised values lead to new values of $\Delta C'_{L1}$, $\Delta C'_{L2}$ and $\Delta C'_{L3}$ from Figures 4 to 6 respectively, given by

$$\Delta C'_{L1} = 0.670,$$
$$\Delta C'_{L2} = 0.970$$

and $$\Delta C'_{L3} = 0.325.$$

Equation (4.20) now gives, with these and values of J_{t1}, J_{t2}, J_{t3} and $(a_1)_0$ from Example 2,

$$\Delta C'_{L0t} = [J_{t1}\,\Delta C'_{L1} + J_{t2}\,\Delta C'_{L2} + J_{t3}\,\Delta C'_{L3}](a_1)_0/2\pi$$
$$= [0.921 \times 0.670 + 1.8 \times 0.970 + 1.42 \times 0.325] \times 5.62/2\pi$$
$$= 2.526,$$

which in Equation (3.3), with c'/c = 1.537, gives

$$\begin{aligned}\Delta C_{L0t} &= \frac{c'}{c}\Delta C'_{L0t} \\ &= 1.537 \times 2.526 \\ &= 3.882\,,\end{aligned}$$

which is about 5% greater than the no slat case in Example 2.

(ii) *Contribution to* ΔC_{Lm}:

The change to the value of $\Delta C'_{Lmt}$ calculated in Example 2 likewise arises from the revised values of $\Delta C'_{L1}$, $\Delta C'_{L2}$, $\Delta C'_{L3}$ and c'/c, so that with appropriate reference to Example 2 for the values of other parameters, Equation (4.31) gives

$$\begin{aligned}\Delta C'_{Lmt} &= (1 - c/c')(1 - \sin\delta_{t1})(C_{LmB})_d + K_T K_{t1} J_{t1} \Delta C'_{L1} + K_T K_{t2} J_{t2} \Delta C'_{L2} \\ &\qquad + K_T K_{t3} J_{t3} \Delta C'_{L3} \\ &= (1 - 1/1.537) \times (1 - \sin 10°) \times 1.309 + 2.5 \times 0.528 \times 0.921 \times 0.670 \\ &\qquad + 2.5 \times 0.215 \times 1.8 \times 0.970 + 2.5 \times 0.08 \times 1.42 \times 0.325 \\ &= 0.378 + 0.815 + 0.938 + 0.092 \\ &= 2.223\,,\end{aligned}$$

which, in Equation (3.4) with F_R = 1 and c'/c = 1.537, gives

$$\begin{aligned}\Delta C_{Lmt} &= F_R(c'/c)\Delta C'_{Lmt} \\ &= 1 \times 1.537 \times 2.223 \\ &= 3.417\,,\end{aligned}$$

which is about 10% greater than the no slat case in Example 2.

2. Slat Contributions

For the given slat geometry Item No. 94027 gives (in the notation of that Item)

$$K_0 = 1.35,\ K_e = 1,\ K_g = 1.96,\ \delta_0 = 0.25 \text{ and } [\,\Delta C'_{L0l}\,]_2 = 0.03\,.$$

Figure 10 of the present Item, with $\delta_l^\circ = 40°$, gives*

$$K_l = 0.625\,,$$

which replaces the value that would be given by Figure 1b of Item No. 94027.

* Note that the value of the dimensionless slat gap G_l/c = 0.05/2.5 = 0.02 is within the range of applicability for Figure 10.

These values, together with $c_{el}/c' = 0.120$ and $\delta_l = 0.698$ rad, are then used to evaluate Equations (3.6) and (3.10) of Item No. 94027 to give

$$\Delta C'_{L0l} = -0.078 \text{ and } \Delta C'_{Lml} = 0.713\,,$$

respectively, which by means of Equations (3.7) and (3.12) of Item No. 94027 with $c'/c = 1.537$ and $F_R = 1$, give

$$\Delta C_{L0l} = -\,0.120 \text{ and } \Delta C_{Lml} = 1.096\,.$$

3. Total Values

Equations (3.1) and (3.2) give the total increments as

$$\begin{aligned}\Delta C_{L0} &= \Delta C_{L0l} + \Delta C_{L0t}\\ &= -\,0.120 + 3.882\\ &= 3.76\,,\end{aligned}$$

which compares with 3.68 from Example 2 with no slat deployment, a small increase of about 2%, and

$$\begin{aligned}\Delta C_{Lm} &= \Delta C_{Lml} + \Delta C_{Lmt}\\ &= 1.096 + 3.417\\ &= 4.51\,,\end{aligned}$$

which compares with 3.10 from Example 2 with no slat deployment, an increase of about 45%.

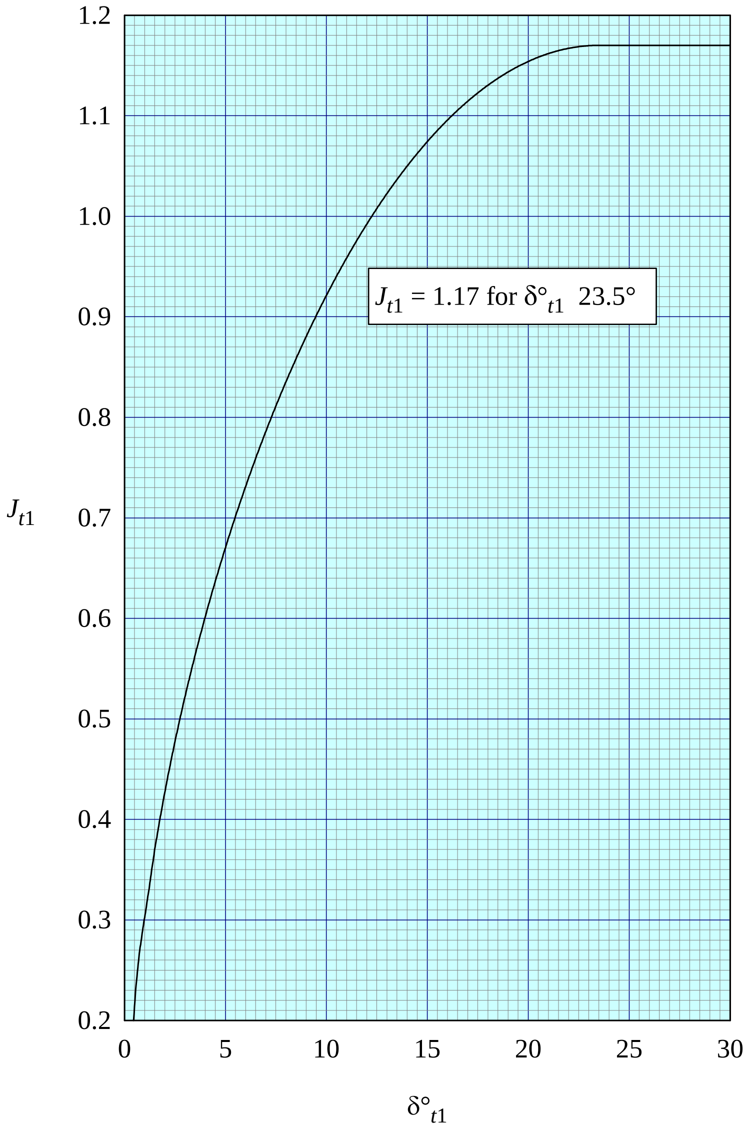


FIGURE 1

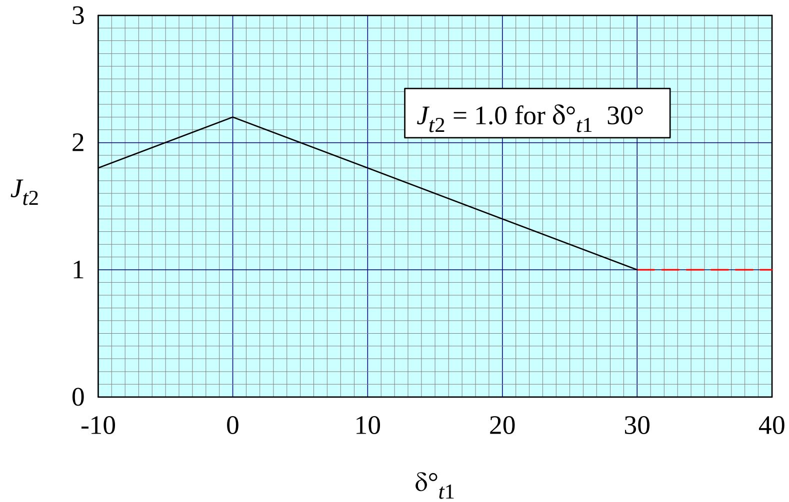


FIGURE 2

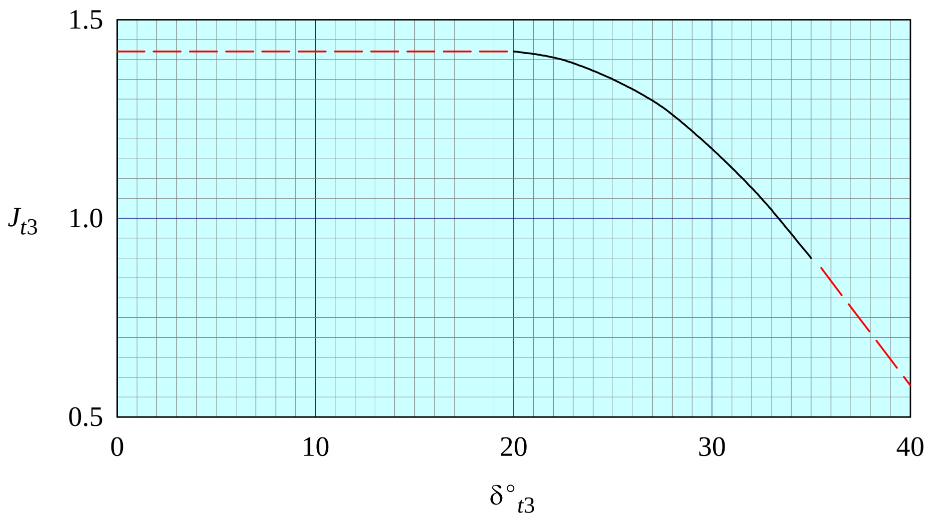
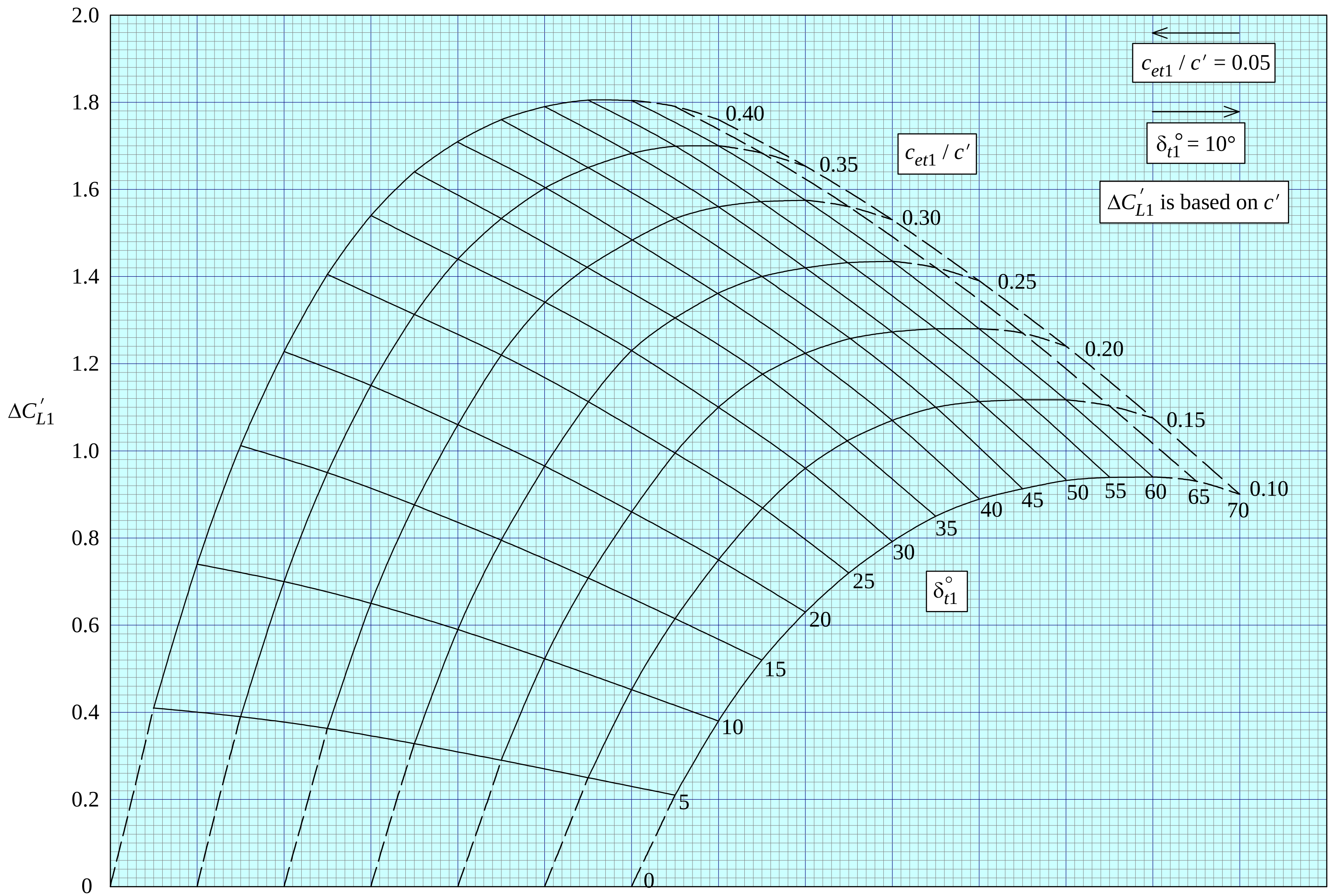


FIGURE 3



$\Delta C'_{L1}$
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0
$c_{et1} / c' = 0.05$
$\delta^{\circ}_{t1} = 10°$
$\Delta C'_{L1}$ is based on c'
c_{et1} / c'
0.40
0.35
0.30
0.25
0.20
0.15
0.10
δ°_{t1}
0
5
10
15
20
25
30
35
40
45
50
55
60
65
70

FIGURE 4

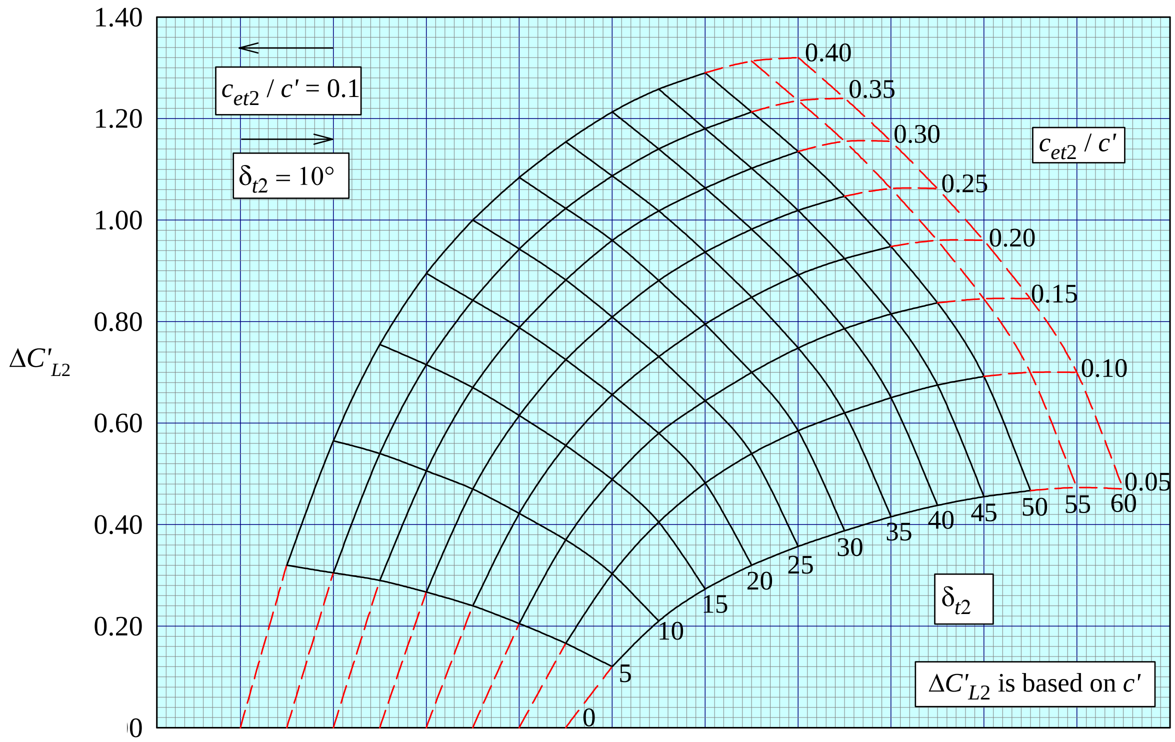


FIGURE 5

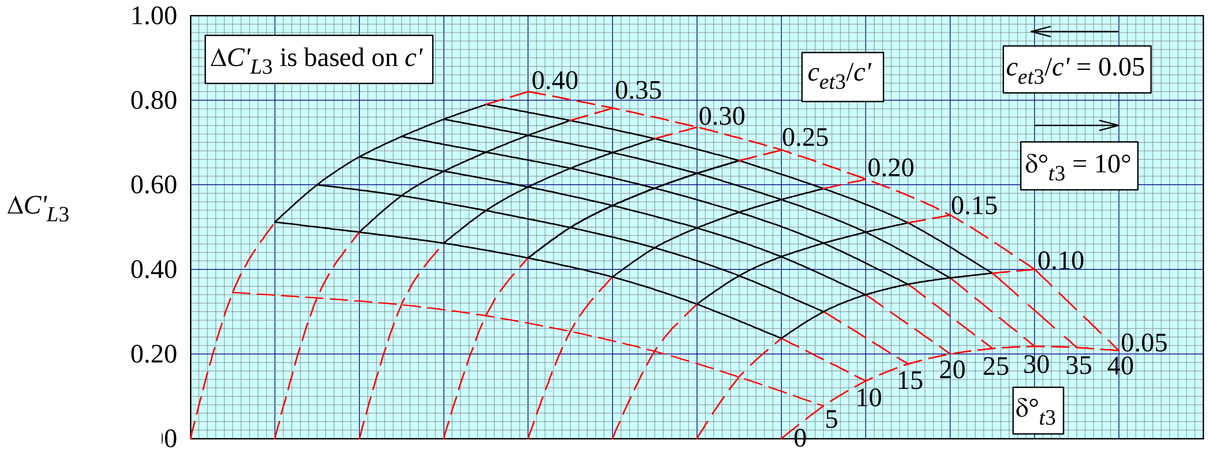


FIGURE 6

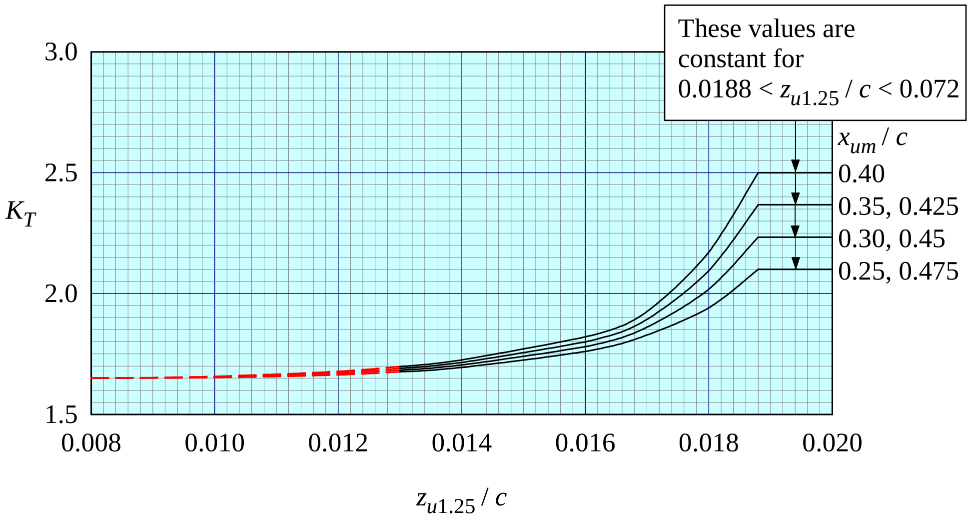


FIGURE 7

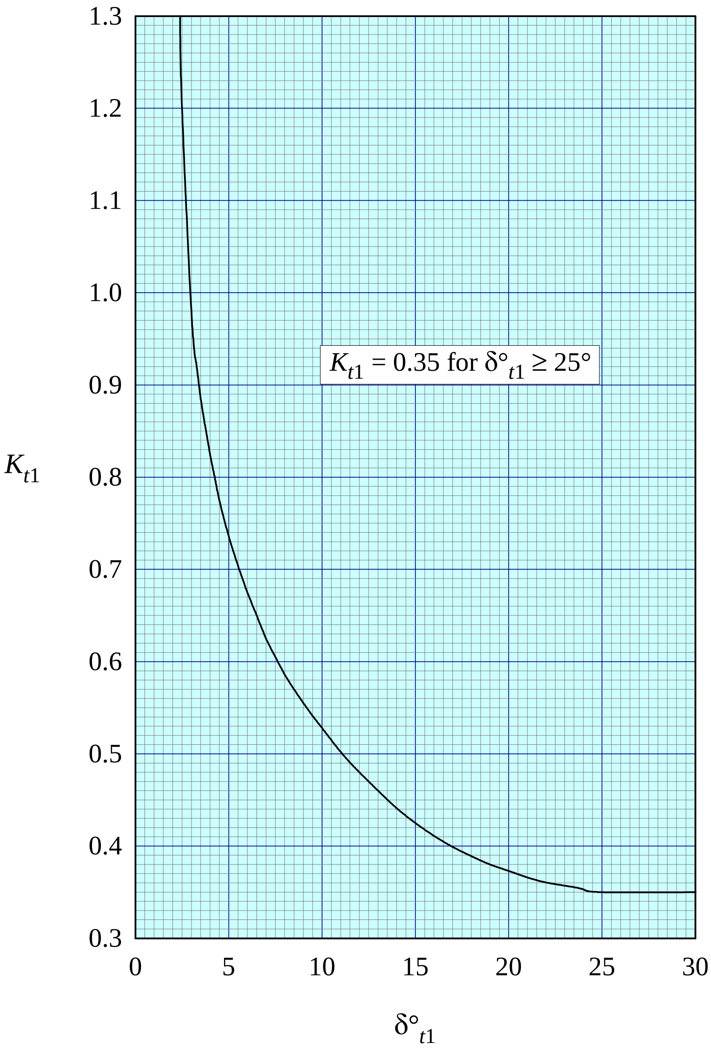


FIGURE 8

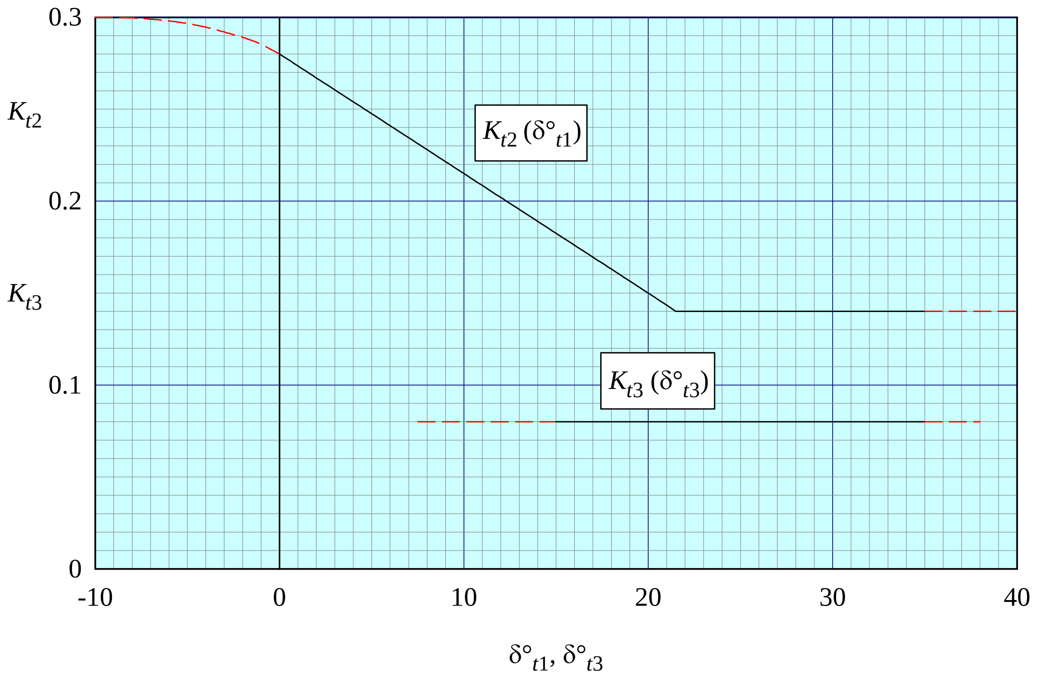


FIGURE 9

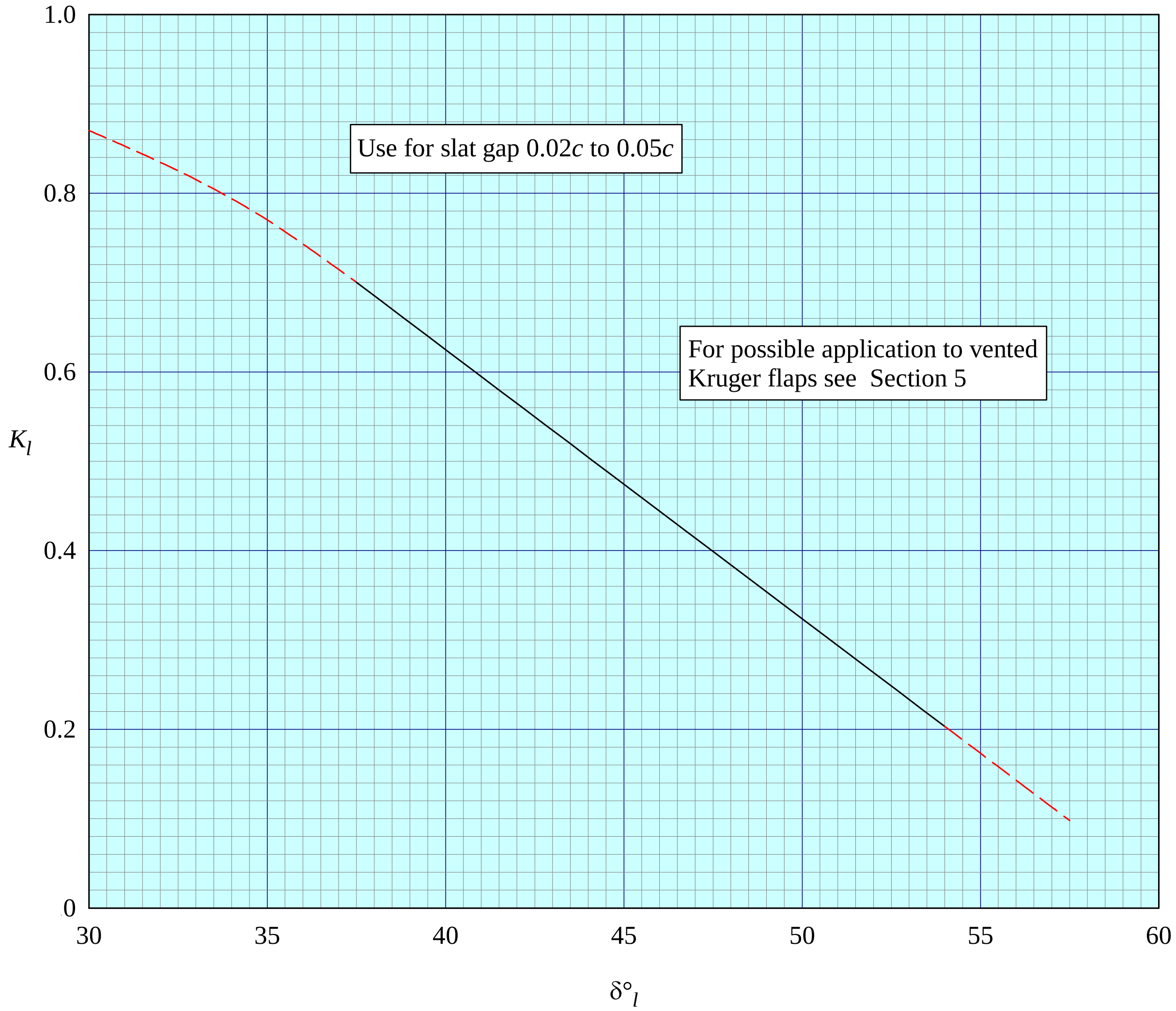


FIGURE 10

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ESDU 94031

Increments in aerofoil lift coefficient at zero angle of attack and in maximum lift coefficient due to deployment of a double-slotted or triple-slotted trailing-edge flap, with or without a leading-edge high-lift device, at low speeds
ESDU 94031

ISBN 978 0 85679 918 1, ISSN 0141-397X

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ESDU 94031 presents an estimation method based on the theory for a thin hinged plate modified using empirical correlation factors to account for the geometry of practical aerofoils and high-lift devices. Some allowance for the effects of chord extension was made by using flap chord ratio and lift coefficients based on aerofoil extended chord but further adjustments were required to adapt to the considerable departure for slotted flaps from the model, possibly involving large chord extensions. The data for aerofoils with trailing-edge flaps deployed from which the methods were developed were extracted from wind-tunnel tests reported in the literature covering a wide range of practical geometries. Fewer data were available for aerofoils with both leading- and trailing-edge devices deployed. The methods apply to Reynolds numbers greater than a million and freestream Mach numbers less than 0.2. The predicted and test data for the lift coefficient increments correlated to within 15 per cent. The use of the methods is illustrated by worked examples. To obtain results for an aerofoil with both leading-edge devices and slotted flaps deployed, ESDU 84026 is used in conjunction with this document and ESDU 94027.